

Data and group coursework

SMM048 (2025-26) Group Coursework 01 - Group 07

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1 Factor Analysis

An exploratory factor analysis (EFA) was conducted on the 25 personality items measured on a 6-point Likert scale. Missing values (NA) were handled using listwise deletion. In addition, several negatively worded items (X1: empathy, X9: carelessness, X10: idleness, X11: quietness, X12: aloofness, X22: avoidance, and X25: superficiality) were reverse-coded so that higher values consistently represented higher levels of the underlying traits.

To determine the appropriate number of factors, parallel analysis, principal component analysis (PCA), and the scree plot in Figure 1 were examined. The scree plot shows a sharp decline in eigenvalues for the first components (5.13, 2.75, 2.14, 1.85, 1.55, and 1.07). According to the Kaiser criterion (eigenvalue > 1), up to six components could be retained. However, the sixth eigenvalue is only slightly above the threshold, and the scree plot levels off after the fifth component, suggesting that additional components contribute little to explaining the overall variance. Considering these results, a five-factor solution was selected. For completeness, the comparison between the five- and six-component solutions obtained using PCA is provided in the Appendix.

After determining the number of factors, the factor loading matrices were estimated and rotated to improve interpretability. Figure 2 presents the rotated factor loading matrices obtained using both varimax and promax rotations. The two rotation methods produce broadly similar loading patterns, suggesting that the underlying factor structure is relatively stable. In most cases, items load strongly on a single factor, while cross-loadings across multiple factors remain limited. The promax rotation allows correlations between factors and produces slightly clearer clustering of items, although the overall interpretation of the factor structure remains consistent across the two rotation methods.

The uniqueness values reported in Table 1 ranges from 0.27 to 0.83, with an average of approximately 0.58. This indicates that the extracted factors explain a moderate proportion of the variance in the observed items. The rotated factor loadings reveal a clear five-factor structure consistent with widely recognised personality dimensions. Factor 1 loads strongly on items related to emotional instability (X16–X20), suggesting that this factor represents neuroticism. Factor 2 reflects extraversion, with items describing sociability and interpersonal engagement (X11–X15). Factor 3 corresponds to conscientiousness, capturing traits such as organisation, diligence, and careful work behaviour (X6–X10). Factor 4 represents agreeableness, including items related to empathy, cooperation, and concern for others (X2–X5). Finally, Factor 5 appears to capture openness-related characteristics, including curiosity, imagination, and intellectual engagement (X21–X23, X25). Overall, the factor analysis results provide empirical support for a five-factor personality structure within the dataset.

$$\begin{aligned}
 \hat{f}_1: \text{emotional instability} &\approx 0.82x_{16} + 0.79x_{17} + 0.71x_{18} + 0.56x_{19} + 0.52x_{20} \\
 \hat{f}_2: \text{extraversion} &\approx 0.59x_{11} + 0.67x_{12} + 0.49x_{13} + 0.61x_{14} + 0.49x_{15} \\
 \hat{f}_3: \text{conscientiousness} &\approx 0.53x_6 + 0.62x_7 + 0.55x_8 + 0.65x_9 + 0.57x_{10} \\
 \hat{f}_4: \text{agreeableness} &\approx 0.60x_2 + 0.66x_3 + 0.45x_4 + 0.58x_5 \\
 \hat{f}_5: \text{openness} &\approx 0.52x_{21} + 0.45x_{22} + 0.61x_{23} + 0.51x_{25}
 \end{aligned}$$

Table 3 presents the rotated factor loadings obtained from the exploratory factor analysis, showing the relationship between the observed variables and the five latent factors. Larger absolute loading values indicate a stronger relationship between an observed variable and the corresponding latent factor. Using item X16 as an example, the loading for Factor 1 is 0.82, while the loadings on the other factors are close to zero, indicating that this item is primarily associated with Factor 1. This suggests that X16 strongly reflects the latent dimension represented by Factor 1. Since Factor 1 is characterised by items associated with emotional instability (X16–X20), it can be interpreted as representing the neuroticism dimension.

$$X_{16} = 0.82\text{Factor1} + 0.09\text{Factor2} - 0.04\text{Factor3} - 0.21\text{Factor4} - 0.08\text{Factor5} + e_{16}$$

1.1 Figures and tables for Q1

Parallel analysis suggests that the number of factors = 6 and the number of components = 5

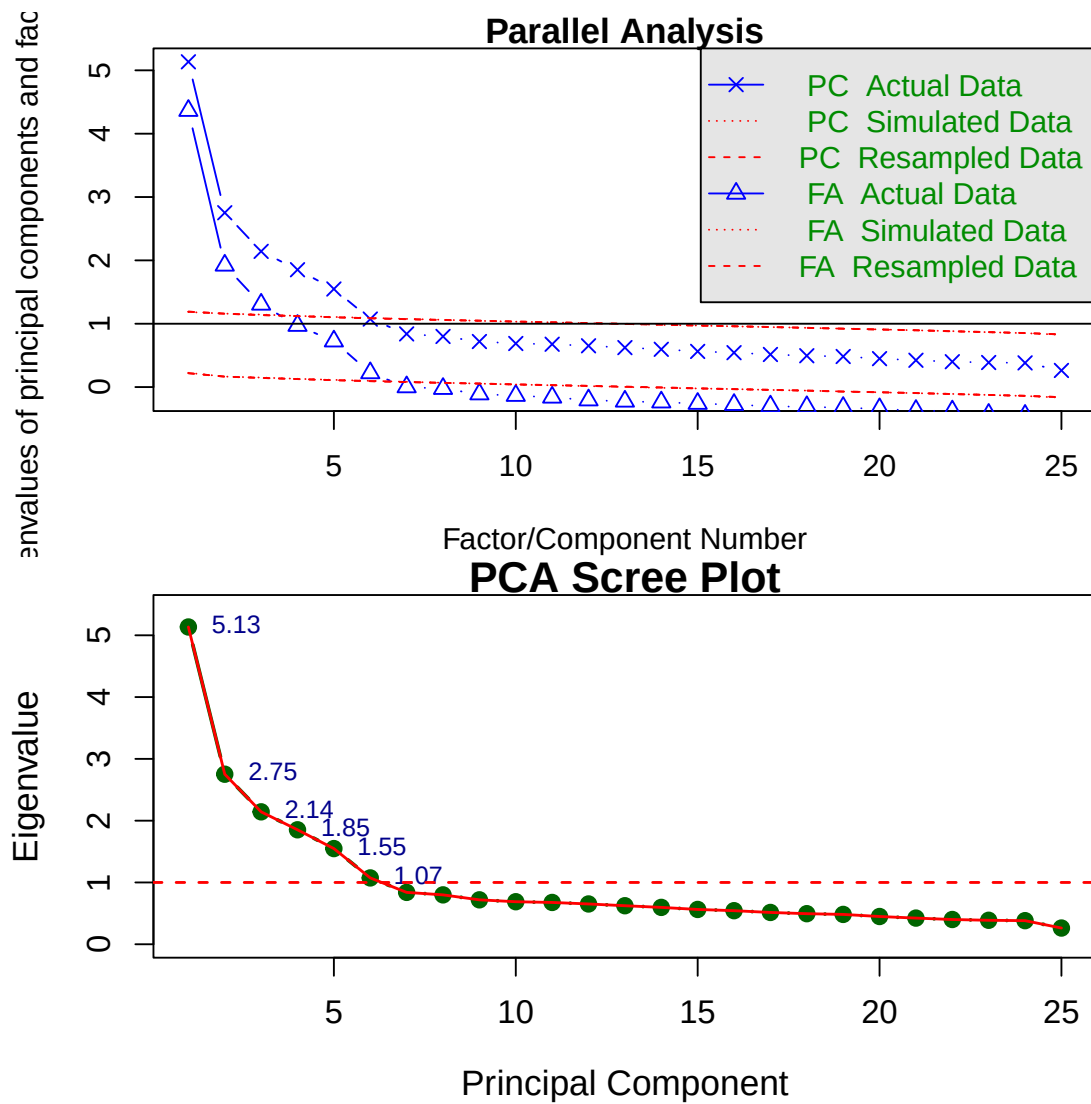


Figure 1: parallel analysis and Scree plot for determining the number of factors

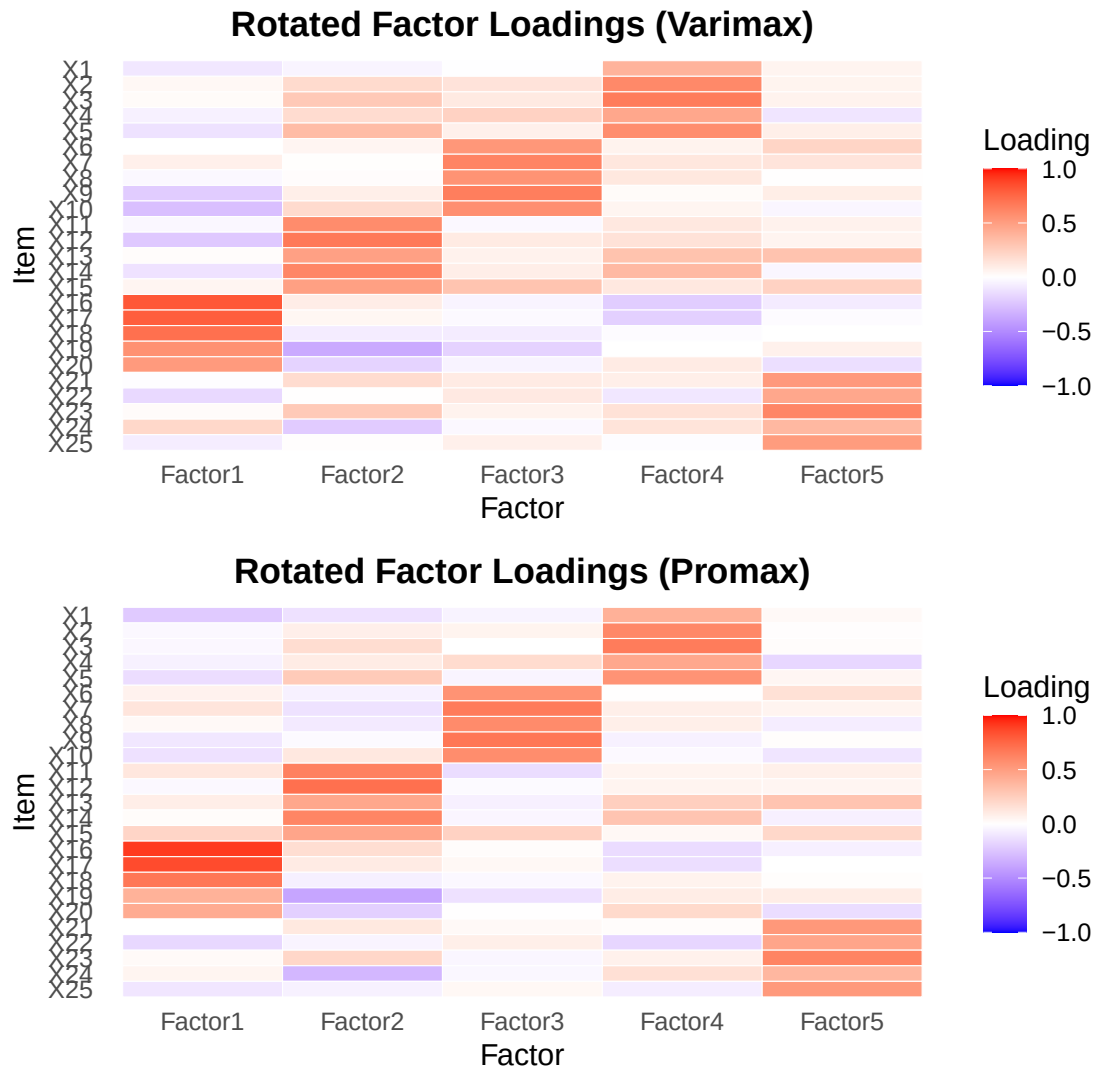


Figure 2: Rotated Factor loadings(Varimax vs Promax)

Table 1: Summary of Uniqueness Values

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.271	0.507	0.569	0.577	0.675	0.83

Table 2: Rotated Factor Loadings for 5 Factors (|loading| ≥ 0.40)

Factor	X1	X2	X3	X4	X5	X6	X7	X8	X9	X10	X11	X12	X13	X14	X15	X16	X17	X18	X19	X20	X21	X22	X23	X24	X25
F1																0.82	0.79	0.71	0.56	0.52					
F2											0.59	0.67	0.49	0.61	0.49										
F3						0.53	0.62	0.55	0.65	0.57															
F4		0.60	0.66	0.45	0.58																				
F5																					0.52	0.45	0.61		0.51

Table 3: Rotated Factor Loadings

	Factor1	Factor2	Factor3	Factor4	Factor5
X1	-0.10	-0.05	0.00	0.39	0.06
X2	0.04	0.19	0.14	0.60	0.06
X3	0.02	0.28	0.11	0.66	0.06
X4	-0.06	0.18	0.23	0.45	-0.11
X5	-0.12	0.35	0.08	0.58	0.08
X6	0.00	0.05	0.53	0.06	0.22
X7	0.08	0.01	0.62	0.13	0.14
X8	-0.03	0.01	0.55	0.12	0.00
X9	-0.22	0.08	0.65	0.02	0.09
X10	-0.27	0.19	0.57	0.05	-0.04
X11	-0.03	0.59	-0.03	0.12	0.07
X12	-0.23	0.67	0.11	0.15	0.06
X13	0.02	0.49	0.07	0.31	0.31
X14	-0.12	0.61	0.09	0.36	-0.04
X15	0.05	0.49	0.31	0.12	0.23
X16	0.82	0.09	-0.04	-0.21	-0.08
X17	0.79	0.04	-0.02	-0.20	-0.02
X18	0.71	-0.08	-0.08	-0.02	0.00
X19	0.56	-0.37	-0.19	0.00	0.07
X20	0.52	-0.19	-0.05	0.11	-0.14
X21	-0.01	0.18	0.10	0.09	0.52
X22	-0.16	0.00	0.11	-0.10	0.45
X23	0.02	0.28	0.07	0.15	0.61
X24	0.21	-0.22	-0.03	0.14	0.37
X25	-0.08	0.01	0.08	-0.01	0.51

2 Copula analysis

As the majority of claim amounts are zero in the dataset (approximately 93.2% of 67,856 records), two approaches were initially considered. The first approach retains all observations and models claim amounts using a Tweedie distribution, which accommodates mixed distributions with probability mass at zero. However, this approach could not be implemented because the current GJRM library does not support joint regression combining Poisson frequency with Tweedie severity margins. The second approach truncates zero-claim observations and models strictly positive claim amounts using alternative continuous distributions. This is also consistent with actuarial practice, where the focus is typically on modelling positive claims for pricing and reserving purposes. The remainder of Question 2 therefore adopts the second approach.

Several truncated count models were considered to account for the absence of zero claims in the dataset, as shown in table 4. The **Truncated Poisson** model is preferred, having the lowest BIC and therefore the most parsimonious. Although the truncated Poisson–Inverse Gaussian specification produces the smallest AIC, the improvement is negligible ($\Delta\text{AIC} \approx 1.13$). The selected model assumes that claims occur independently at a constant rate over the exposure period, with an estimated annual claim rate of $\hat{\mu} = e^{-2.66711} = 0.0694$ indicating that claims arrive relatively infrequently in the portfolio.

Claim severity was modelled using positive claim amounts only. Distribution is strongly right-skewed, indicating a small proportion of claims have extremely high severity. Among the models tested, the **Inverse Gaussian** distribution provides the best fit, achieving the lowest AIC and BIC values, as shown in table 5. It has an average positive claim severity of approximately $\hat{\mu} = e^{7.60808} = 2011$, with variability increasing with the cube of the mean, consistent with the heavy right tail commonly observed in severity data.

As an extension, we have also attempted to include covariates to model the marginal distribution parameters, but we did not see much improvement in the model AIC and BIC. For more details, please refer to `Adding covariates into the models` section in the appendix.

Both graphical and statistical methods are used for model diagnostic. The residual quantile Q–Q plots show that the truncated Poisson model provides a satisfactory representation of claim frequency, while the inverse Gaussian distribution captures the overall shape of the severity distribution. Although the lower tail appears to deviate from the diagonal line, indicating that the model underestimates very small claim amounts, the overall fit is considered adequate for modelling dependence in the following copula analysis. [Refer to residual quantile QQ plot of marginal distributions fits here]. Convergence diagnostics also indicate stable estimation. In both marginal models the maximum absolute gradient is close to zero and the observed information matrix is positive definite, confirming that the likelihood optimisation has successfully converged. Additional goodness-of-fit tests analysis we have done is discussed in `Goodness-of-fit Tests` section in the appendix.

To model the dependence between claim frequency and claim severity, four copula families were considered: Gaussian, Student-t, Clayton and Gumbel. The results are presented in table 6. Among these candidates, the Clayton copula provides the best fit, achieving the lowest AIC and BIC values by a clear margin. Convergence diagnostics confirm stable estimation, with a maximum gradient of 8.45×10^{-6} and a positive definite information matrix. The estimated Clayton copula parameter is positive and statistically significant $\theta = 1.47$, indicating positive dependence between claim frequency and claim severity. It corresponds to a Kendall's $\tau = \frac{\theta}{(\theta+2)} = 0.4236$, which indicates a moderate positive association between claim frequency and claim severity. We therefore fit **Clayton copula** with `claytonCopula(param = 1.47, dim = 2)`.

The Clayton copula captures lower-tail dependence, meaning that policies with relatively few claims tend to generate relatively small claim costs. This is reflected in the simulated scatter plot, where observations cluster in the lower-left region of the unit square. Such asymmetric dependence cannot be captured effectively by symmetric copulas such as the Gaussian or Student-t, which impose identical dependence in both tails. The lower performance of the Gumbel copula in the previous section suggests weaker upper-tail dependence, meaning that extremely high claim counts are not systematically associated with extremely large claim amounts. This may reflect the heterogeneity of severe claims, where large losses can arise from factors such as accident type or medical costs rather than the number of claims alone.

2.1 Figures and tables for Q2

Table 4: Claim frequency model performance

Models	df	AIC	BIC
Truncated Poisson	1	2297.331	2303.771
Truncated NB I	1	2296.227	2309.105
Truncated Poisson Inverse Gaussian	2	2296.119	2309.077

Table 5: Claim severity model performance

Models	df	AIC	BIC
LogNormal	2	77708.31	77721.19
Gamma	2	79329.84	79342.72
Inverse Gaussian	2	77187.65	77200.53

Table 6: Copula model performance

Models	df	AIC	BIC
Gaussian	4	79384.86	79410.61
Student-t	4	79590.55	79616.30
Clayton	4	79321.95	79616.30
Gumbel	4	79482.90	79508.66

3 Principal Component Analysis

We applied Principal Component Analysis (PCA), an unsupervised dimensionality-reduction method, to transform the 11 variables into a smaller set of uncorrelated components that successively explain the maximum possible variance. Variables were standardised using z-scores before PCA because features are in different units, e.g., income vs age vs rating scores. Without scaling, high-variance variables can dominate PC1 purely due to measurement scale rather than the real underlying structure in the data.

The first three principal components explain approximately 55% of the total variance, which is sufficient to summarise the main behavioural patterns in the data. Parallel analysis was also examined to determine the appropriate number of components. Although the test suggests retaining two components, the third component still contributes meaningful interpretability and increases the cumulative variance explained. Therefore, three components were retained for further interpretation.

Parallel analysis suggests that the number of factors = 3 and the number of components = 3

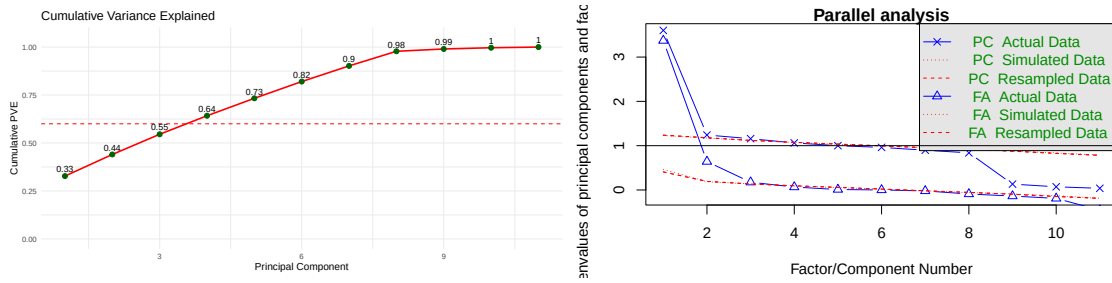


Figure 3: Cumulative variance, parallel analysis.

The rotated loading plot suggests that RC1 is mainly driven by purchasing behaviour, with income, brand loyalty, shopping frequency, and spending score loading strongly on this component, indicating a dimension of purchasing intensity. RC2 is primarily associated with age and spending score, indicating a contrast between younger high-spending customers and older customers, noting that the sign of loadings is arbitrary. RC3 captures variation in marketing channel engagement: promotion usage and email engagement load positively, whereas social media engagement loads negatively, indicating different preferences across communication channels. Overall, the three rotated components summarise customer behaviour into purchasing intensity, demographic influence, and marketing responsiveness, providing a simplified representation of the main behavioural patterns in the data.

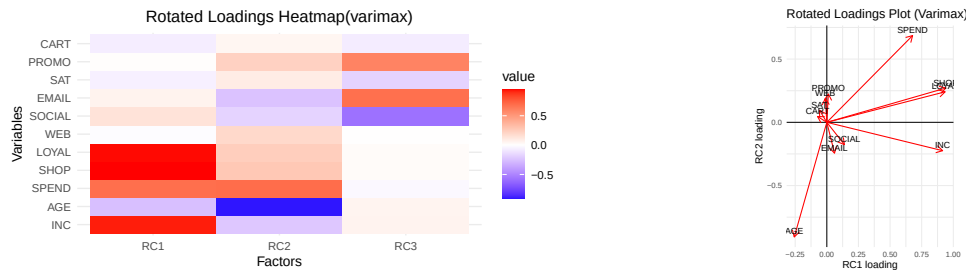


Figure 4: Heatmap and loading plot of the rotated PCA loadings using varimax rotation.

Table 7: Rotated PCA loadings after varimax rotation ($|\text{loading}| \geq 0.40$)

Factor	INC	AGE	SPEND	SHOP	LOYAL	WEB	SOCIAL	EMAIL	SAT	PROMO	CART
RC1	0.92		0.68	0.94	0.94						
RC2		-0.91	0.69								
RC3							-0.58	0.67		0.59	

4 Time series

4.1 Description of time series:

We first plotted the three time series (ts1, ts2 and ts3) and visually assessed the stationarity of each. The plots for ts1 (5) and ts2 suggest they are stationary as they exhibit constant mean and variances, without trend nor seasonality. However, the plot for ts3 suggests non-stationarity with evidence of trend. Formal stationarity test such as Augmented Dickey–Fuller (ADF) tests confirmed these: the p-values for ts3 is large (0.7322), so the null hypothesis of non-stationarity cannot be rejected, whereas the p-values for ts1 and ts2 are smaller than 0.01.

4.2 Model selection

Model selection heuristics:

1. Use auto.arima to pick p, d, q parameters as a baseline
2. Manually fit Arima models for possible combinations of p and q (from 0 to 5 inclusive each), and consider the top 5 models ranked by AIC and BIC.
3. Check residuals for models with smaller BIC or AIC than the baseline. The most parsimonious model with satisfactory residuals is then selected.

The final models selected for each time series is:

TS1: (2,0,1) is selected over (2,0,2) suggested by auto.arima because of better AIC and BIC, i.e better fit and simpler.

TS2: (2, 0, 3) as suggested by auto.arima is selected because other models with smaller BIC such as (0, 0, 2), (1, 0, 2) and (1, 0, 2) all exhibit serial correlation according to Ljung-Box tests.

TS3: (2, 1, 0) from auto.arima already has the best AIC and BIC.

4.3 Model Diagnostics

T1: The diagnostic plots for the fitted ARIMA(2,0,1) model for ts1 are shown in Figure 6. The residual ACF and PACF display no significant spikes outside the confidence bounds, suggesting that no substantial autocorrelation remains in the residuals. The Q–Q plot in Figure 7 indicates that the residuals are approximately normally distributed. In addition, the standardized residuals fluctuate randomly around zero with roughly constant variance. Overall, these diagnostics suggest that the ARIMA(2,0,2) model provides an adequate fit for ts1, as illustrated in Figure 8.

T2: The residual diagnostics indicate that most autocorrelations lie within the confidence bounds and the standardized residuals fluctuate randomly around zero. Furthermore, the Q–Q plot in Figure 10 suggests that the residuals are approximately normally distributed, and overall the diagnostics indicate that the fitted model provides an adequate description of ts2, as shown in Figure 11.

T3: The Q–Q plot in Figure 13 suggests that the residuals are approximately normally distributed. Furthermore, as shown in Figure 14, the standardized residuals fluctuate randomly around zero and the residual ACF shows no significant autocorrelation. The Ljung–Box test results also support the conclusion that the residuals behave like white noise, indicating that the fitted model provides an adequate description of the series.

4.4 Figures and tables for Q4

Table 8: Top 5 ARIMA models for time series 1 ranked by AIC

p	q	AIC	BIC
2	3	550.860	573.948
2	4	552.701	579.087
3	3	552.819	579.206
4	2	552.890	579.277
5	5	553.855	593.435

Table 9: Top 5 ARIMA models for time series 1 ranked by BIC

p	q	AIC	BIC
2	1	554.022	570.513
1	2	554.119	570.611
1	1	559.434	572.627
2	3	550.860	573.948
2	2	555.449	575.239

Table 10: Top 5 ARIMA models for time series 2 ranked by AIC

p	q	AIC	BIC
2	4	637.204	663.591
4	5	637.619	673.901
3	5	637.663	670.646
0	2	638.011	651.204
3	4	638.479	668.163

Table 11: Top 5 ARIMA models for time series 2 ranked by BIC

p	q	AIC	BIC
0	2	638.011	651.204
1	2	639.797	656.289
0	3	639.816	656.308
2	2	641.316	661.106
0	4	641.493	661.283

Table 12: Top 5 ARIMA models for time series 3 ranked by AIC

p	q	AIC	BIC
2	0	555.135	565.015
2	5	556.704	583.051
2	1	556.830	570.003
3	0	556.851	570.024
5	2	557.587	583.934

Table 13: Top 5 ARIMA models for time series 3 ranked by BIC

p	q	AIC	BIC
2	0	555.135	565.015
2	1	556.830	570.003
3	0	556.851	570.024
3	1	557.974	574.441
0	3	561.945	575.119

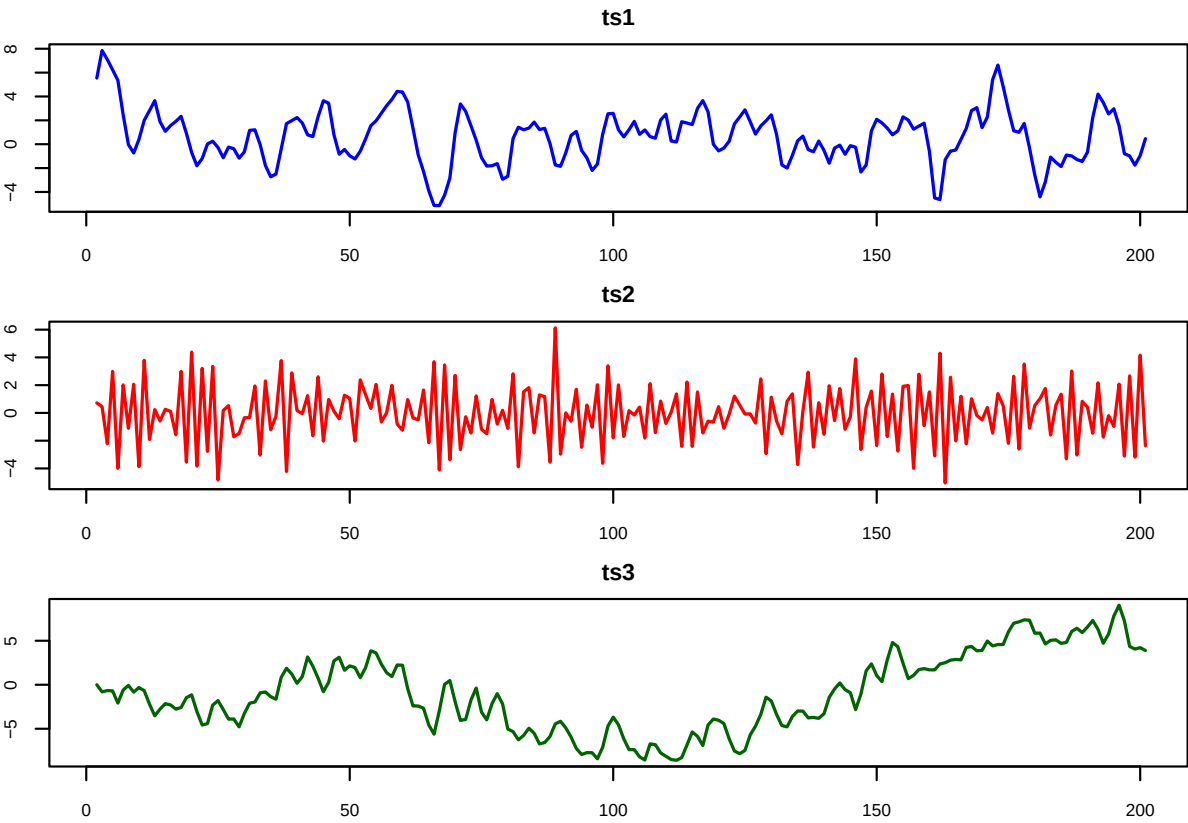
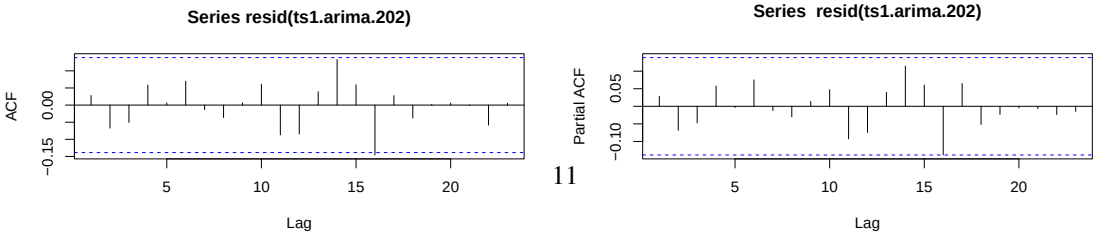


Figure 5: Time plots of the three simulated series

Table 14: ADF tests and suggested ARIMA models

Series	ADF_statistic	p_value	ARIMA_model
ts1	-6.050	0.010	2
ts2	-4.108	0.010	0
ts3	-1.627	0.732	2
ts1	-6.050	0.010	2
ts2	-4.108	0.010	0
ts3	-1.627	0.732	3
ts1	-6.050	0.010	2
ts2	-4.108	0.010	1
ts3	-1.627	0.732	0



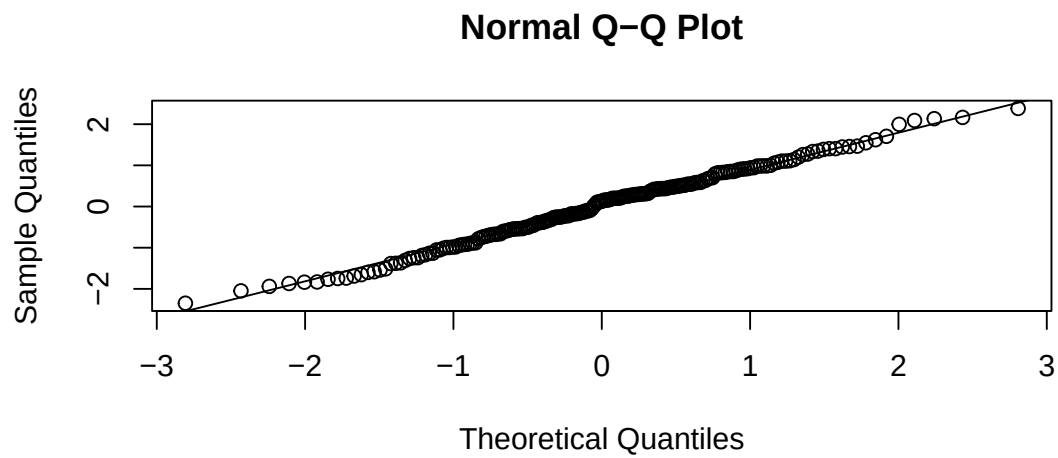


Figure 7: QQPLOT

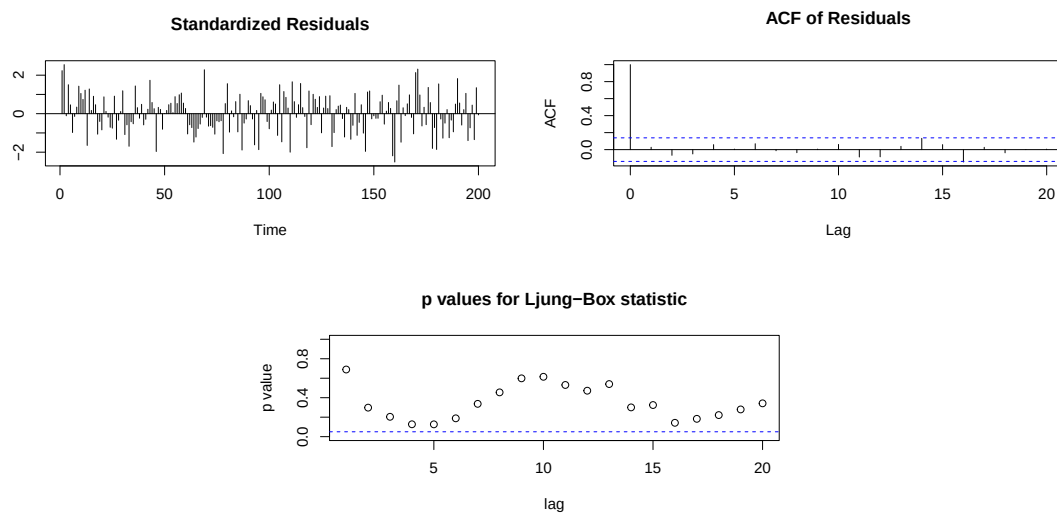


Figure 8: Diagnostic plots for the fitted ARIMA(2,0,2) model for ts1, including standardized residuals, residual ACF, and Ljung-Box test results.

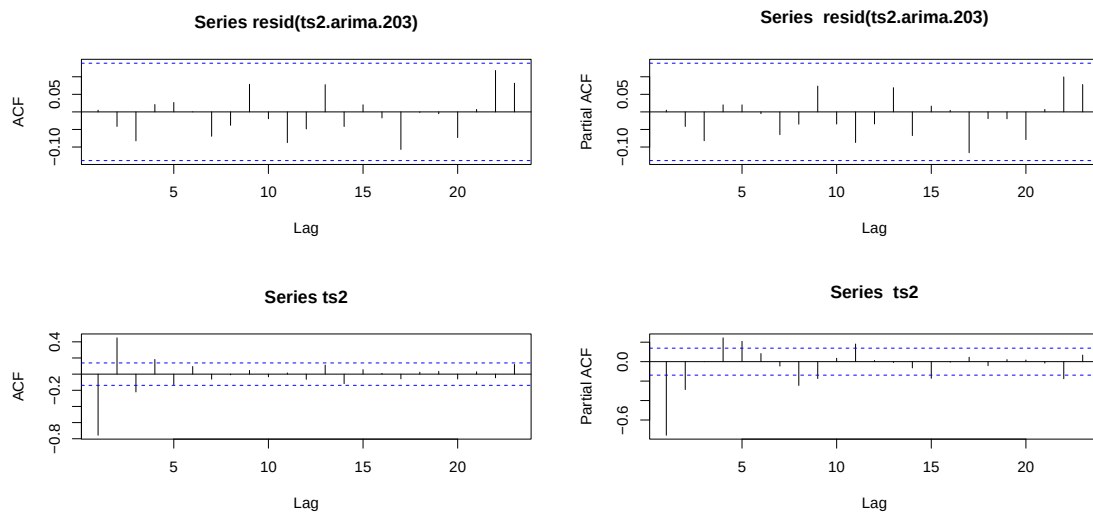


Figure 9: ACF vs PACF

Normal Q-Q Plot

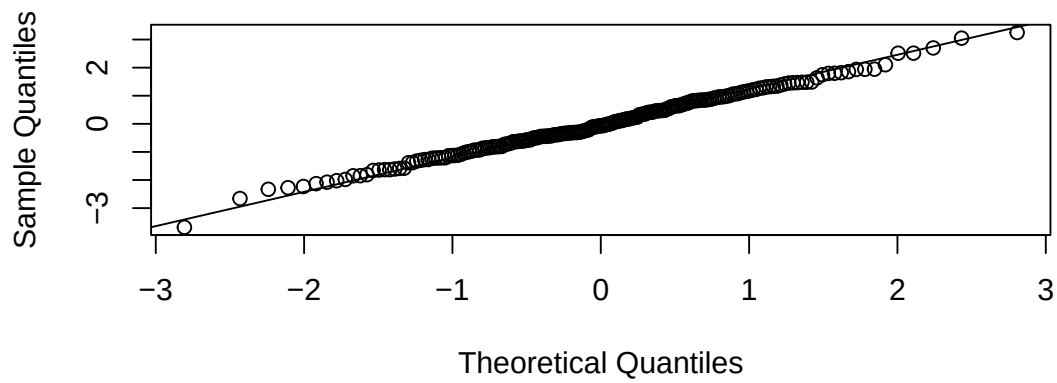


Figure 10: QQPLOT

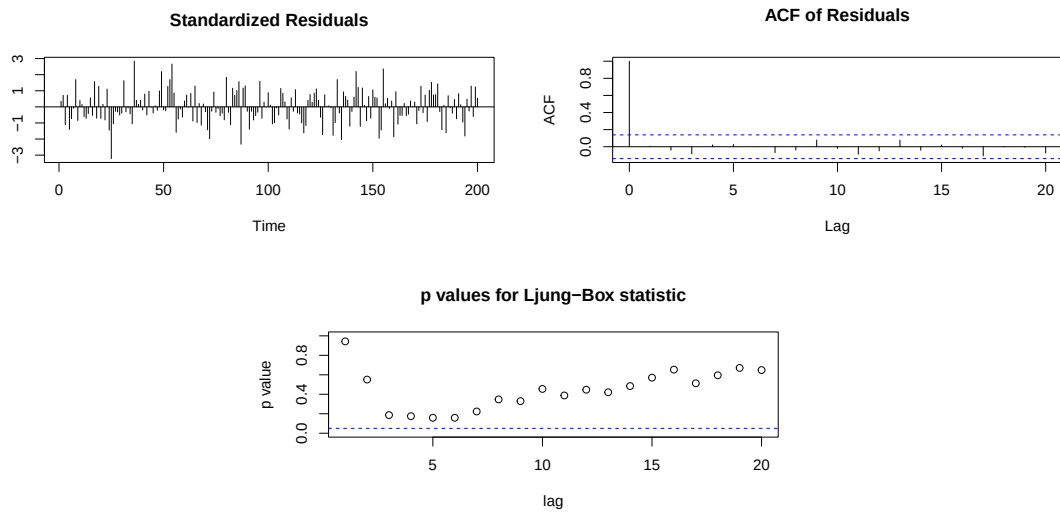


Figure 11: Diagnostic plots for the fitted ARIMA(2,0,3) model for ts_2 , including standardized residuals, residual ACF, and Ljung-Box test results.

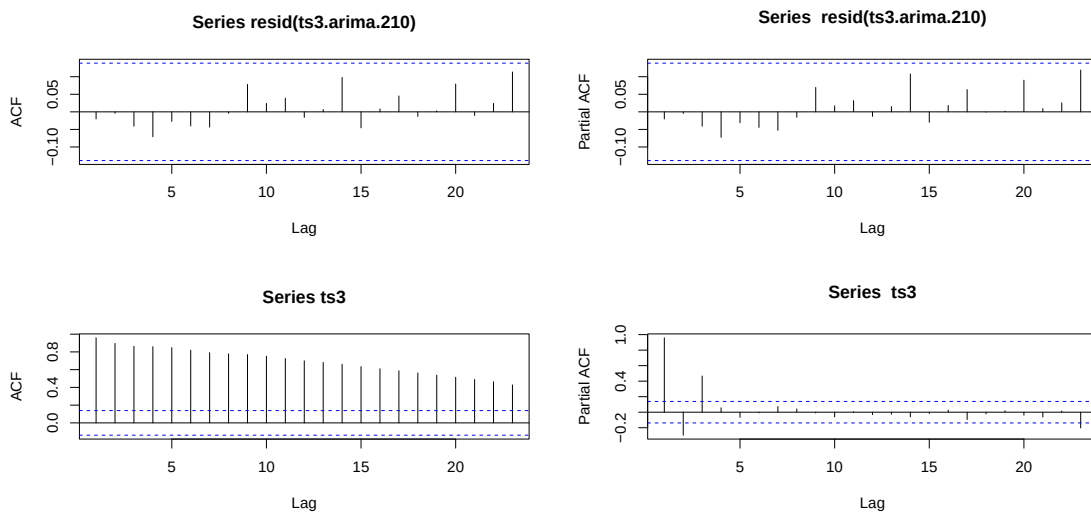


Figure 12: ACF vs PACF

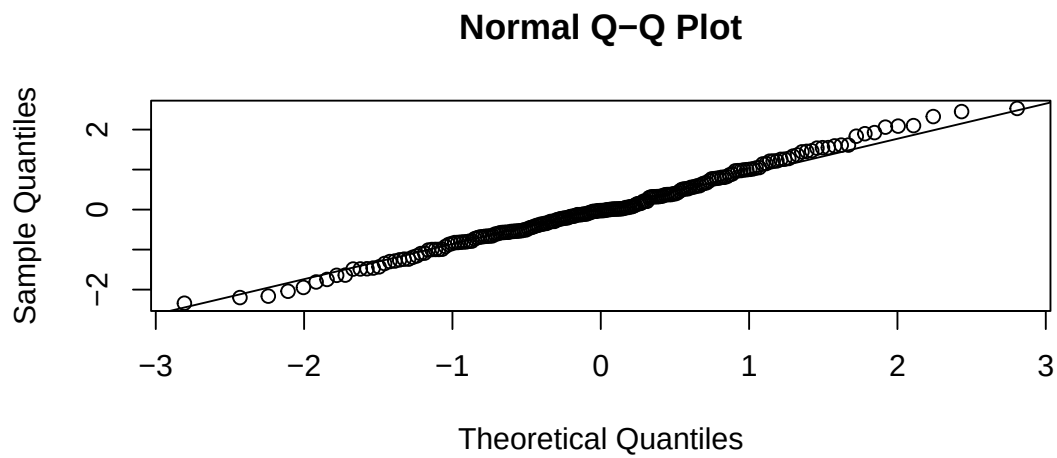


Figure 13: QQPLOT

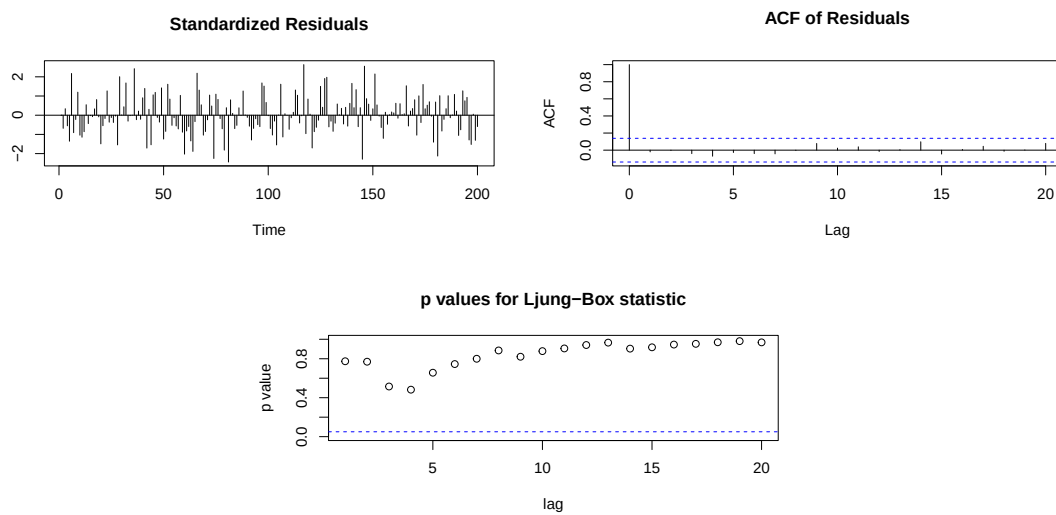


Figure 14: Diagnostic plots for the fitted ARIMA(2,1,0) model for ts3, including standardized residuals, residual ACF, and Ljung-Box test results.

5 Appendix

5.1 bonus : Q1 Factor analysis comparison(number of factors 5 vs 6)

The heatmaps compare the rotated factor loading structures obtained from the five-factor and six-factor solutions using varimax rotation. In the five-factor solution, items generally load strongly on a single factor, producing a relatively clear and interpretable structure. When a sixth factor is introduced, several items show weaker or more dispersed loadings across factors, suggesting a less stable structure. In particular, the additional sixth factor captures only a limited number of items and does not form a clearly interpretable dimension. Therefore, the five-factor solution appears more parsimonious and provides a more coherent representation of the underlying personality traits.

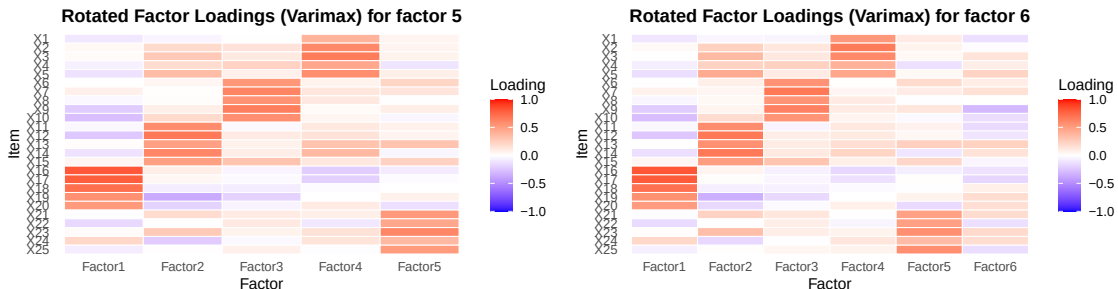


Figure 15: Comparison of the number of factors (5 vs 6)

The rotated factor loading tables allow a comparison between the five-factor and six-factor solutions. In the five-factor model, the items cluster clearly into five interpretable groups, with most salient loadings above 0.40 concentrated on a single factor. In contrast, the six-factor solution does not substantially improve the structure, as the additional sixth factor (F6) does not show any strong loadings above the threshold. Most items continue to load on the same factors as in the five-factor solution, indicating that the extra factor does not capture a distinct dimension. Therefore, the five-factor model appears more parsimonious and provides a clearer and more interpretable representation of the underlying personality structure.

Table 15: Rotated Factor Loadings for 5 Factors ($|\text{loading}| \geq 0.40$)

Factor	X1	X2	X3	X4	X5	X6	X7	X8	X9	X10	X11	X12	X13	X14	X15	X16	X17	X18	X19	X20	X21	X22	X23	X24	X25
F1																0.82	0.79	0.71	0.56	0.52					
F2											0.59	0.67	0.49	0.61	0.49										
F3						0.53	0.62	0.55	0.65	0.57															
F4		0.60	0.66	0.45	0.58																				
F5																					0.52	0.45	0.61		0.51

Table 16: Rotated Factor Loadings for 6 Factors ($|\text{loading}| \geq 0.40$)

Factor	X1	X2	X3	X4	X5	X6	X7	X8	X9	X10	X11	X12	X13	X14	X15	X16	X17	X18	X19	X20	X21	X22	X23	X24	X25
F1																0.82	0.81	0.71	0.56	0.51					
F2					0.44						0.58	0.67	0.57	0.67	0.51										
F3						0.56	0.68	0.55	0.64	0.55															
F4		0.53	0.66	0.60	0.40	0.46																			
F5																					0.48	0.49	0.57		0.57
F6																									

5.2 bonus : Q2 Copula

5.2.1 Goodness-of-fit tests

Goodness-of-fit tests were conducted using the Kolmogorov–Smirnov (KS) and Cramér–von Mises (CvM) statistics. Null hypothesis is data follow the fitted marginal distribution. The KS statistic ($D1=0.93707$, $D2=0.74048$) measures the maximum difference between the empirical and theoretical cumulative distributions of zero truncated poisson and inverse gaussian, while the Cramér–von Mises statistic ($\omega_1^2 = 1268.4$, $\omega_1^2 = 653.09$) measures the overall squared discrepancy across two distributions. Both tests reject the null hypothesis that the observed data follow the fitted distributions exactly ($p\text{-value} < 2.2 \times 10^{-16}$). However, with a sample size of $n = 4624$, these tests are highly sensitive and may detect even minor deviations from the theoretical distribution.

5.2.2 Adding covariates into the models

Additional models including covariates were explored to assess whether observable policy characteristics improve the marginal model specification. In particular, vehicle value, geographic area and gender were considered, as these variables may plausibly influence accident risk and claim costs. For example, vehicle value may affect repair costs, while area and gender may capture differences in driving behaviour or traffic exposure. In the claim frequency model, the intercept-only truncated Poisson specification achieves the lowest AIC and BIC compared to models with extra covariates. For the claim severity model, the inclusion of 4 extra covariates, including vehicle value, area and gender producing the lowest AIC. However, BIC again favours the simpler intercept-only specification. Overall, it is suggested that the additional covariates provide limited explanatory power in this dataset.

[Put table here]

5.3 bonus : Q3 Principal Component Analysis

The correlation matrix shows generally weak to moderate correlations among variables, suggesting that several variables related to purchasing behaviour (spending score, Shopping Frequency, and Brand loyalty) are strongly positively correlated. PC1 contrasts purchasing intensity (spending score, shopping frequency, and brand loyalty) against age, suggesting that older customers tend to have lower purchasing-intensity scores on this component(here can add scatterplot as evidence to support if have space).



Figure 16: Correlation matrix (numbers left, heatmap right)

The scree plot suggests retaining three to four components. Although the Kaiser criterion indicates four components (eigenvalue > 1), the elbow in the scree plot and the interpretability of the rotated loadings support retaining three components.

PCA Scree Plot

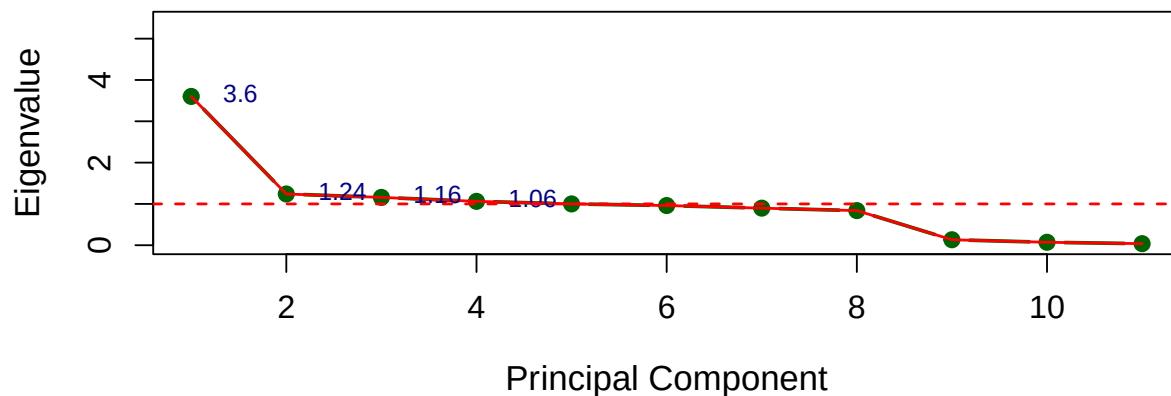


Figure 17: Scree plot

The scatter plot shows the distribution of observations in the space of the first two principal components, with clusters identified using k-means clustering. The results suggest three groups of customers with different behavioural patterns. One group is associated with higher purchasing intensity (high PC1 values), while another group corresponds to lower purchasing activity. The third group lies between these extremes, representing customers with moderate purchasing behaviour.

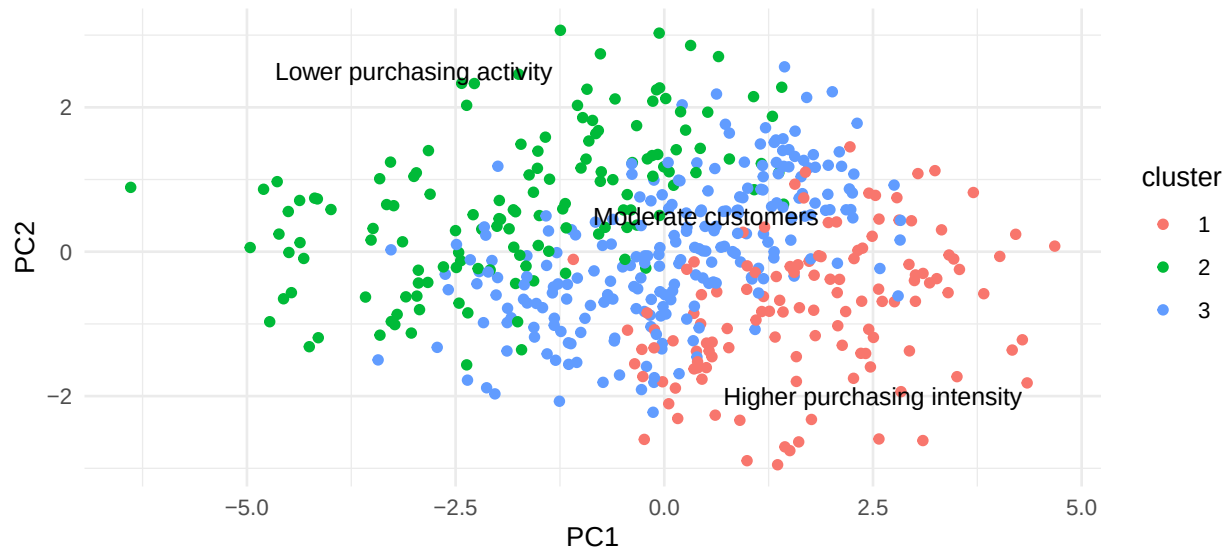


Figure 18: Scatter plot

The PCA biplot illustrates both the observations and the directions of the original variables in the space of the first two principal components. Variables related to purchasing behaviour, such as income, brand loyalty, shopping frequency, and spending score, are largely aligned with the direction of PC1, indicating their strong contribution to this component. In contrast, age is oriented in a different direction, suggesting that it captures a different aspect of variation in the data. Overall, the biplot confirms that purchasing-related variables primarily drive the variation explained by the first principal component.

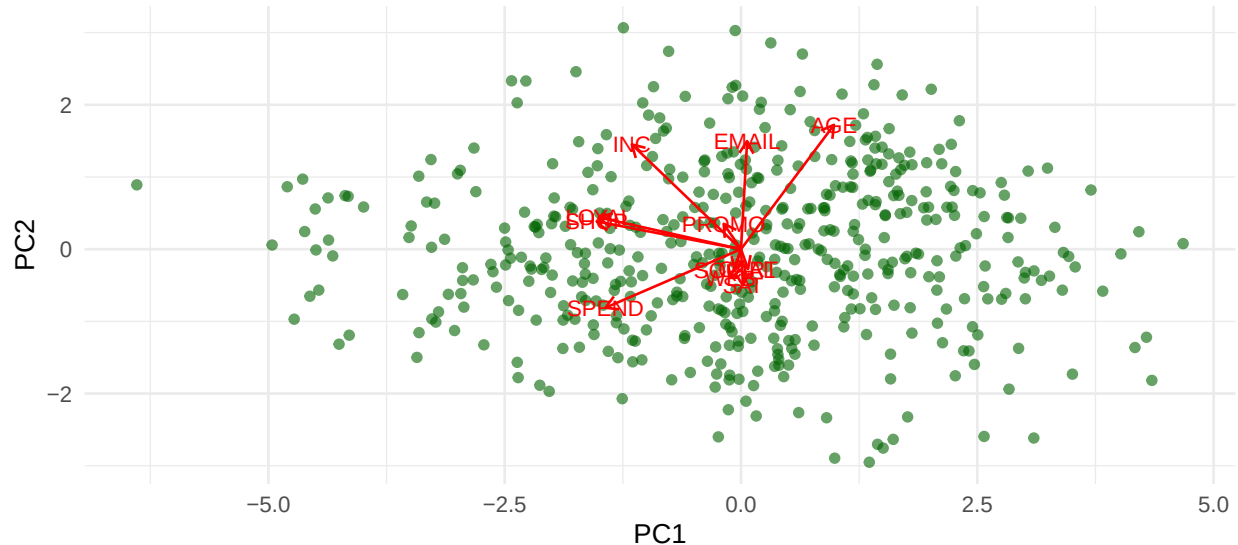


Figure 19: PCA Biplot

The three-dimensional PCA score plot provides a visual representation of the observations across the first three principal components. The points form a continuous distribution rather than clearly separated clusters, indicating that customer behaviour varies gradually across the dataset. This visualization further supports the interpretation that the main variation in the data is captured by the first few principal components.

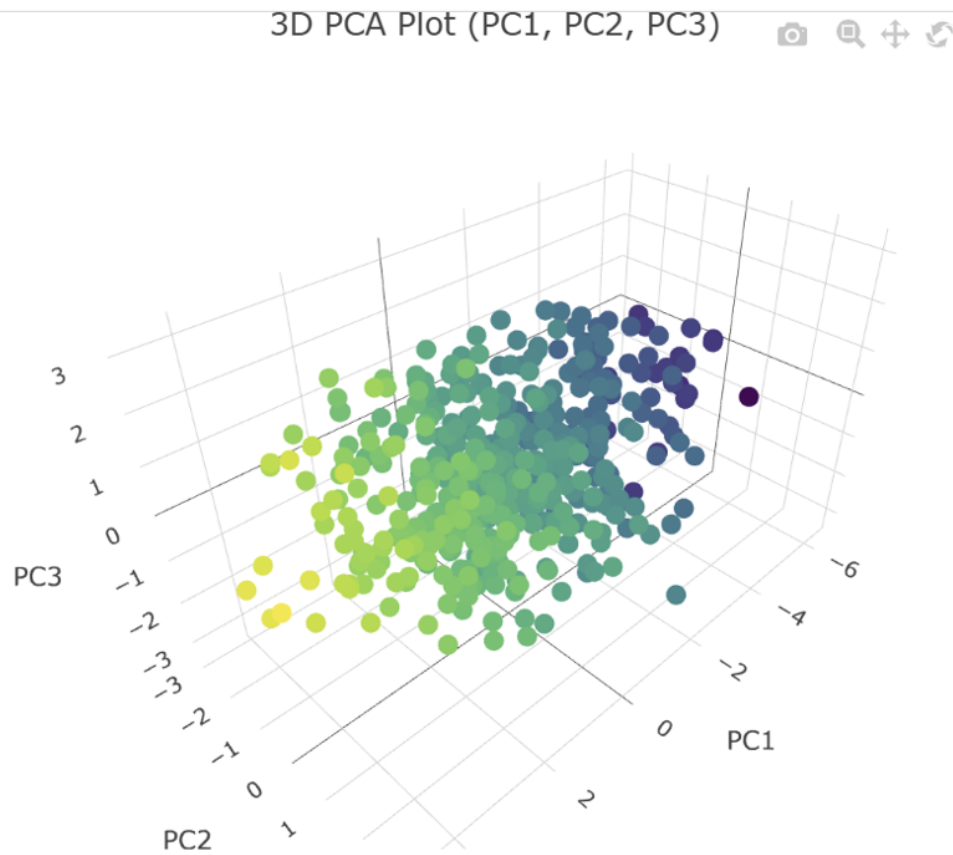


Figure 20: 3D PCA score plot of the first three principal components.

5.4 bonus : Q4 Time series